Exp. No: 08

NMAP - Host Discovery

Aim:

To study Nmap host discovery techniques and identify live hosts in a network using ARP, ICMP, TCP and UDP scans.

Task 1 - Introduction To Nmap

1.) What is Nmap?

*. Nmap (Network Mapper) is a free and open-source network scanning tools used to discover hosts, networks and services.

2.) Who created Nmap?

*. Nmap was created by Gordon Lyon (Fyodor).

3) What are the main purposes of Nmap?

*. To discover live hosts.
*. To identify running services.
*. To scan networks and ports.
*. To perform network reconnaissance.

Task 2 - Subnetworks.

1.) How many devices can see the ARP Request sent from computer1?

*. 4.

2.) Did computer6 receive the ARP Request?

*. Nay.

3) How many devices can see the ARP Request sent from computer4?

*. 4.

4) Did computer6 reply to the ARP Request?

*. Yea.

TASK 3 - HOST DISCOVERY USING TCP/IP

1) What type of packet did computer1 send before the Ping?
   * ARP Request.
2) What type of packet did computer1 receive before sending Ping?
   * ARP Response.
3) How many computers responded to the Ping Request?
   * 1.
4) What was the first device that responded to the second ARP Request?
   * Computer5.
5) Did another Ping Request require new ARP Requests?
   * Nay.

TASK 4 - ENUMERATING TARGETS.

1) What is the first IP address Nmap would scan for 10.10.12.13/29
   * 10.10.12.8
2) How many IP addresses will Nmap scan for 10.10.0.255.101-105?
   * 6400

TASK 5 - Nmap HOST DISCOVERY USING ARP.

1) How many hosts were found alive after scanning CONNECTION_IP/24?
   * 1

TASK 6 - Nmap Host Discovery USING ICMP.

1) What option is used for ICMP Timestamp discovery?
   * -PP.

a) What option is used for TCMP Address Mask discovery?

* -PM

b) What option is used for TCMP Echo discovery?

* -PE

TASK 7 - Nmap Host Discovery Using TCP and UDP

a) Which TCP Ping scan does not require a privileged account?

* TCP SYN Ping

b) Which TCP Ping scan requires a privileged account?

* TCP ACK Ping

c) What option is required to run a TCP SYN Ping scan on the Telnet port?

* -PS23

TASK 8 - REVERSE DNS LOOKUP

a) Which option is used to perform reverse DNS lookup for all hosts?

* -R

TASK 9 - SUMMARY

a) What is the purpose of -sn?

* It performs host discovery only without port scanning.

b) What is the purpose of -n?

* It disabled DNS lookup

c) What is the purpose of -R?

* It performs reverse DNS lookup for all hosts.

d) What option is used for ARP scan?

* -PR

5) What option is used for ICMP Echo Scan?

* -PE

6) What option is used for ICMP Timestamp Scan?

* -PP

7) What option is used for ICMP Address Mask Scan?

* -PM

8) What option is used for TCP SYN Ping?

* -PS

9) What option is used for TCP ACK Ping?

* -PA

10) What option is used for UDP Ping?

* -PU

RESULT:

Thus, Nmap host discovery techniques such as ARP, ICMP, TCP, and UDP were studied and used to identify live hosts in a network.